Quantitative Crypto Trading Case Study

Cullen Trace Campana

Contents

1 Executive Summary **3**
1.1 Core Strategy: Altcoin Pairs Trading . . . 3
1.2 Approach and Design Choices . . . 3
1.3 Risks . . . 4
1.4 Path Forward . . . 4

2 Data Acquisition **5**
2.1 Data Collection and Venue Selection . . . 5
2.2 Data Collection Methodology . . . 5
2.2.1 Altcoin Data Collection (Binance) . . . 5
2.2.2 Data Storage and Standardisation . . . 5
2.3 Quality Assessment and Cross-Validation . . . 5
2.3.1 Data Validation . . . 5
2.3.2 Known Limitations and Gaps . . . 6
2.4 Technical Challenges Overcome . . . 6
2.4.1 Binance Universe Stability . . . 6

3 Backtesting Framework **7**
3.1 Altcoin Pairs Backtesting Engine . . . 7
3.1.1 Architecture Overview . . . 7
3.1.2 Data Handling . . . 7
3.1.3 Execution Model . . . 7
3.1.4 Transaction Cost Model . . . 8
3.1.5 Margin and Leverage . . . 8
3.1.6 Equity Tracking and Performance Measurement . . . 8
3.1.7 Validation Approach . . . 9

4 Altcoin Pairs Strategy Implementation **10**
4.1 Universe Construction . . . 10
4.2 Multi-Stage Screening Pipeline [2] . . . 10
4.2.1 Stage 1 – Return Correlation Pre-Filter . . . 11
4.2.2 Stage 2 – Engle-Granger Cointegration Test . . . 11
4.2.3 Stage 3 – Half-life Filter . . . 11
4.2.4 Stage 4 – ADF + KPSS Conjunction Stationarity Test . . . 11
4.2.5 Stage 5 – Variance Ratio Test . . . 11
4.2.6 Stage 6 – Spread Distribution Filter . . . 11
4.2.7 Stage 7 – Structural Break Detection . . . 12
4.2.8 Stage 8 – Benjamini-Hochberg FDR Correction . . . 12
4.2.9 Pair Ranking and Selection . . . 12
4.3 Signal Generation and Kalman Filter . . . 12
4.3.1 Kalman Filter for Hedge Ratio Estimation . . . 12
4.3.2 EWMA Z-Score Signal . . . 13
4.4 Position Sizing and Risk Management . . . 13
4.4.1 Volatility-Adjusted Leg Sizing . . . 13
4.4.2 Implied Correlation Regime Filter . . . 14
4.4.3 Confidence-Based Sizing . . . 14
4.4.4 Exit Cascade . . . 14

4.4.5 Additional Risk Controls . . . 15
4.5 Strategy Performance . . . 15
4.5.1 Headline Results . . . 15
4.5.2 Trade Quality Analysis . . . 16
4.5.3 Rolling Sharpe Analysis . . . 17
4.6 Crisis Performance Analysis . . . 18
4.6.1 UST/Luna Collapse (May–June 2022) . . . 18
4.6.2 FTX Bankruptcy (November–December 2022) . . . 18
4.6.3 USDC Depeg / Banking Crisis (March 2023) . . . 18
4.6.4 SEC Lawsuits – Binance and Coinbase (June–July 2023) . . . 19
4.6.5 Summary . . . 19
4.7 Capacity and Sensitivity . . . 19
4.7.1 Capacity Estimate . . . 19
4.7.2 Parameter Sensitivity . . . 19
4.8 Deviations from Case Study Specification . . . 19
4.9 Enhancements Beyond Specification . . . 20
4.10 Future Improvements . . . 22
4.10.1 Strategy Enhancements . . . 22
4.10.2 Venue Expansion . . . 23

A Data Acquisition Plan **25**
A.1 Data Requirements Matrix . . . 25
A.2 Source Evaluation . . . 25
A.3 Decision Criteria . . . 25
A.4 Collection Strategy . . . 26
A.4.1 Technical Approach . . . 26
A.4.2 Rate Limiting & Error Handling . . . 26
A.4.3 Storage Format . . . 26
A.4.4 Timeline and Risk Mitigation . . . 26
A.4.5 Fallback Plans . . . 26

B Data Quality Report **28**
B.1 Coverage Analysis . . . 28

1 Executive Summary

This report presents the design, implementation, and backtested performance of a quantitative cryptocurrency trading framework centred on altcoin statistical arbitrage. Rather than spreading effort thinly across a broad specification, I chose to build one strategy to production depth, a Kalman-filter-based pairs trading engine with a multi-stage cointegration pipeline, novel regime filtering, and a look-ahead-proof backtesting architecture. The result is a strategy that has been stress-tested across 6,690 trades and four major crypto crises over four years, with the level of engineering rigour required for live deployment.

1.1 Core Strategy: Altcoin Pairs Trading

The altcoin pairs strategy operates on Binance 15-minute candles across a dynamic universe of approximately 140 actively traded tokens. Pairs are selected through a nine-stage screening pipeline with Benjamini–Hochberg false discovery rate control ($q = 0.001$), and hedge ratios are estimated via a Kalman filter with MLE-fitted process noise, replacing the standard OLS approach with a method that is both theoretically superior and computationally more efficient. Position sizing integrates three independent confidence signals: Kalman posterior uncertainty, an implied correlation regime filter (a novel decomposition of the hedge ratio into correlation and volatility components), and volatility-adjusted leg sizing for variance-neutral exposure.

The strategy was profitable through all four major crypto crises in the sample period. Market-driven panic events (UST/Luna collapse, FTX bankruptcy, USDC depeg) actually enhanced performance, the Luna period alone generated $169K in P&L at a 67.6% win rate, because crisis-driven selling creates widespread temporary dislocations between cointegrated pairs. The November 2025 BTC liquidation event produced the only significant drawdown (∼10%), from which the strategy recovered within weeks.

The backtesting engine enforces look-ahead prevention structurally, not by convention: the `DataFeed` emits one bar at a time, making it impossible for the strategy to access future data. Transaction costs are modelled at institutional-tier rates (0.04% maker, 0.06% taker, 0.01% slippage), and a full cross-margin simulation tracks leverage, interest, and liquidation risk, though peak actual leverage never exceeded 1.65× despite a 3× allowance, demonstrating that the confidence-based sizing naturally self-regulates exposure.

1.2 Approach and Design Choices

I deliberately chose depth over breadth: rather than implementing multiple strategy components at a surface level, I invested the majority of effort into building the altcoin pairs strategy to a standard where the screening pipeline, signal generation, risk management, and backtesting infrastructure could support live trading with minimal additional engineering.

This choice involved tradeoffs. I did not implement a DEX universe for the pairs strategy (DEX execution costs of 0.5–1.5% round-trip would consume the strategy's per-trade edge at the current holding period). These are documented transparently in the deviations table in Section 4.9.

The rationale was straightforward: a single strategy with 6,690 trades, a 1.45 Sharpe, and survival through four crises provides stronger evidence of genuine edge than multiple strategies with thin backtests and ambiguous statistical significance.

1.3 Risks

Five principal risks apply to the framework as presented:

Regime dependence is the most material risk. The rolling Sharpe analysis reveals clear regime shifts, with the strategy performing best during range-bound markets and worst during violent directional moves. Implementing an explicit regime detector to modulate position sizing is the single highest-priority improvement.

Transaction cost sensitivity is inherent in a high-frequency mean-reversion strategy. Cumulative costs totalled $1.56M on $1M initial capital, any loss of institutional fee tier would materially impair net returns. Cost-aware entry gating (estimating expected P&L per trade and rejecting entries where costs dominate) would directly address this.

Single-venue concentration means that a Binance-specific event (regulatory action, API outage, withdrawal freeze) would halt all trading. The codebase supports venue-agnostic cost models; extending execution to Bybit or OKX is an engineering task, not a research problem.

Regulatory risk manifested directly in the backtest: the SEC lawsuits period (June–July 2023) produced the weakest crisis performance, with more structural pair breaks and a lower win rate than market-driven crises. Regulatory events selectively impact tokens in ways that statistical models cannot anticipate.

Capacity constraints limit the current strategy to approximately $3M without optimisation. Expanding to more pairs, incorporating order-book liquidity data, and adding secondary venues could extend this to $10M+.

1.4 Path Forward

The largest marginal improvement to this framework is not further model refinement, the screening pipeline and Kalman filter are already sophisticated, but expanding the data universe. Persistent pricing inefficiencies exist on less crowded venues, and these inefficiencies are likely even more pronounced for altcoin pairs, where DEX liquidity pools price tokens differently from CEX order books. Incorporating data from additional venues into the existing pairs framework, which was designed from the start to be venue-agnostic, would broaden the opportunity set substantially while trading on venues where competition is thinner and alpha is more accessible.

In hindsight, the data collection effort was disproportionately focused on Binance, a highly competitive, efficient exchange where alpha is hardest to find. The real mispricings lie on less crowded exchanges. The framework is ready for them; the data pipeline needs to catch up.

2 Data Acquisition

2.1 Data Collection and Venue Selection

For the altcoin statistical arbitrage strategy, Binance was selected as the sole data source. Binance dominates global spot trading volume for USDT pairs, and strategy execution is assumed to occur on Binance. Consequently, Binance order book and trade data are the most relevant inputs for realistic backtesting.

A dynamic universe of the top 100 most liquid USDT pairs was constructed on a monthly basis. Liquidity ranking was based on average daily quote volume. To mitigate survivorship bias while preserving liquidity constraints, a sticky carry-over mechanism was implemented: assets that fell out of the top 100 were retained in the universe provided they maintained a minimum average daily volume of $2 million.

This design balances breadth of coverage with survivorship bias awareness. Importantly, dropped symbols are explicitly tracked and logged with reason codes (e.g., low volume or delisting), ensuring transparency in universe construction.

2.2 Data Collection Methodology

Binance provides a free, public API requiring no authentication. The collection pipeline is implemented in a Python script targeting the altcoin data domain.

2.2.1 Altcoin Data Collection (Binance)

The script `binance_top100_15m.py` collects altcoin data via the Binance REST API using a two-phase process.

First, a ranking phase evaluates all USDT trading pairs based on their 7-day average quote volume. Assets are ranked monthly to construct the top-100 liquidity universe. Second, a download phase retrieves 15-minute OHLCV candles for the active universe.

The script supports resumable execution through a `--skip-existing` flag and persists the evolving universe state to CSV files between runs. This ensures robustness to interruptions and transparency in universe construction over time.

2.2.2 Data Storage and Standardisation

All outputs are stored in Apache Parquet format and organised by venue. A consistent schema is enforced across datasets, including harmonised field naming and explicit tagging of each record with its `venue` and `venue_type`. This standardisation enables seamless downstream strategy implementation.

2.3 Quality Assessment and Cross-Validation

2.3.1 Data Validation

The collection script includes built-in validation routines executed after data collection.

These routines report:

- Start and end date ranges

- Gap analysis (median and maximum inter-candle gaps)
- Price range statistics

This automated validation layer ensures internal consistency, detects missing segments, and highlights anomalous values prior to strategy backtesting.

2.3.2 Known Limitations and Gaps

The primary limitation of the dataset is the absence of DEX-native data. All altcoin data originates from Binance, and no on-chain DEX data (e.g., Uniswap, Curve) was collected. This was a deliberate scope decision. The strategy under analysis does not require DEX execution, and collecting high-quality multi-chain DEX data would introduce substantial complexity, including subgraph pagination, wash-trading filtering, and liquidity threshold enforcement, with limited incremental benefit for the backtesting exercise.

For a production system, DEX integration would be a natural extension, particularly for price validation and long-tail token discovery.

Additionally, no market capitalisation, on-chain metrics, or sentiment data were collected. The strategy relies exclusively on price and volume data. Incorporating protocol-level metrics (e.g., TVL, fee revenue, large wallet flows) from sources such as DefiLlama or Dune Analytics would strengthen potential machine learning extensions.

2.4 Technical Challenges Overcome

2.4.1 Binance Universe Stability

Constructing a consistent altcoin universe across multiple years required careful handling of token listings, delistings, and liquidity regime shifts. The sticky universe design, whereby assets are retained as long as they remain above a minimum volume threshold (even after falling out of the top 100), reduces excessive turnover and provides greater stability for cointegration analysis.

Explicit logging of dropped symbols, along with reason codes, enables downstream survivorship bias diagnostics.

3 Backtesting Framework

3.1 Altcoin Pairs Backtesting Engine

3.1.1 Architecture Overview

The backtesting engine is designed around a single overriding principle: look-ahead bias must be structurally impossible, not merely avoided by convention. The architecture enforces this through a strict separation of concerns across four components: a `DataFeed` that emits one bar at a time, a `RollingWindow` that accumulates historical context without exposing future data, a `PairsEngine` that processes each bar through a deterministic pipeline, and a Portfolio that tracks positions and computes equity.

At each 15-minute bar, the engine executes the following pipeline in order: (1) update the EWMA volatility tracker for all assets, (2) push the current bar into the rolling window, (3) if a refit is due, screen for cointegrated pairs and reinitialise the Kalman filter, (4) force-close any positions whose constituent assets have missing data, (5) batch-update the Kalman filter to produce spread estimates and z-scores, (6) evaluate the exit cascade for all open positions, (7) evaluate entry logic for new signals, and (8) check margin and liquidation conditions. The strategy code never has access to the full price matrix, only to the current bar and the contents of the rolling window, which by construction contains only past observations.

3.1.2 Data Handling

The `DataFeed` ingests a multi-asset `DataFrame` of 15-minute OHLCV candles sourced from Binance. Internally, all prices are converted to log space for numerical stability. Log prices make addition and subtraction equivalent to percentage returns, which simplifies spread computation and avoids distortions that arise when operating on raw prices for assets that differ by orders of magnitude (e.g., BTC at $\sim$ \$50,000 versus a small-cap altcoin at \$0.10). The `DataFeed` yields a fresh copy of each row to the strategy, preventing any accidental write-back into the internal price matrix.

The `RollingWindow` maintains a circular buffer of configurable depth (6,000 bars, approximately 62 days). When the strategy requests historical data for a given asset, the window returns a contiguous chronological array with NaN entries for bars where the asset was unavailable. This design handles the reality of a dynamic universe where coins are listed and delisted throughout the backtest period.

3.1.3 Execution Model

All trades are assumed to execute at the close price of the bar on which the signal is generated. This is a reasonable approximation for a 15-minute bar strategy on Binance spot, where liquidity for the assets in our universe (minimum \$5–10M daily volume) is sufficient to fill orders without meaningful slippage at the position sizes I consider. The strategy could further reduce market impact by scaling into positions over multiple bars, though this is not modelled explicitly.

Each pair trade consists of two legs: a long position in one asset and a short position in the other, with leg sizes determined by the volatility-adjusted sizing model described in Section 4.4.1 Entry and exit for both legs are assumed to occur simultaneously at the same bar.

3.1.4 Transaction Cost Model

Transaction costs are modelled through a venue-aware cost framework (`costs.py`) that separates proportional costs (fees + slippage) from fixed costs (gas, where applicable). For the Binance venue used in this backtest, the cost parameters are:

- Maker fee: 0.04% (4 bps)
- Taker fee: 0.06% (6 bps)
- Estimated slippage: 0.01% (1 bp)
- Gas: $0.00 (CEX, no on-chain settlement)

A round-trip pair trade involves four legs (open leg A, open leg B, close leg A, close leg B), giving a total round-trip cost of approximately 0.12% of notional. These parameters reflect institutional-tier fee schedules available to market makers and high-volume traders on Binance. The assumed slippage of 1 bp is conservative for the liquidity levels in our universe and could be achieved by using limit orders or by scaling into positions gradually rather than executing market orders on a single bar.

Over the full backtest, cumulative transaction costs totalled $1,555,434 on $1M initial capital. This figure reflects the strategy's high turnover (6,690 trades over approximately 4 years, with a median holding period of 25 bars or roughly 6 hours). While the cost drag expressed as a percentage of initial capital appears large (155%) (this is a consequence of leverage and high turnover), when measured as a fraction of total notional traded, the cost per trade is consistent with the assumed fee schedule.

3.1.5 Margin and Leverage

The engine includes a full cross-margin simulation (`margin.py`) that models Binance spot cross-margin mechanics. The margin configuration for this backtest was set at 3x maximum leverage with a 5% maintenance margin rate, 2% liquidation fee, and hourly interest charges of 0.03% on borrowed amounts.

In practice, the strategy never approached its leverage limit. Peak actual leverage reached 1.65x, with peak margin utilisation of 55%. The conservatism arises naturally from the position sizing model: each pair is allocated a maximum of 5% of equity, and with up to 30 pairs active, the theoretical maximum gross exposure is 150% of equity, well within the 3x limit. Zero liquidation events occurred during the backtest period. Total interest paid on borrowed amounts was $4,361, a negligible fraction of total P&L.

3.1.6 Equity Tracking and Performance Measurement

Equity snapshots are recorded every four bars (one hour), capturing total equity, realised and unrealised P&L, transaction costs, number of open positions, and margin state. Performance metrics are computed from daily-resampled equity to avoid Sharpe inflation that arises from sub-daily return series (where many zero-return periods dilute the standard deviation and $\sqrt{N}$ annualisation assumes i.i.d. returns, which fails for strategies that are frequently flat). Sharpe and Sortino ratios are computed against a 4% risk-free rate.

3.1.7 Validation Approach

The backtest runs as a continuous rolling-window simulation over the full period from January 2022 to December 2025. No in-sample/out-of-sample split or walk-forward optimisation is applied. This is a deliberate choice: the strategy's pair selection and Kalman filter parameters are re-estimated from scratch at every refit (approximately every 7.5 days), and the screening pipeline's restrictive statistical thresholds ($p < 0.001$ with Benjamini–Hochberg FDR correction at $q = 0.001$) are designed to minimise overfitting risk at each refit.

The rolling nature of the backtest means that pairs are continuously entering and exiting the active universe based on their current statistical properties, providing a form of built-in out-of-sample validation, the strategy can only trade pairs that pass screening on live (unseen) data at each refit.

4 Altcoin Pairs Strategy Implementation

4.1 Universe Construction

Deviation from case study specification:

The case study specifies a dual-universe construction with separate CEX and DEX components. Our implementation uses a CEX-only universe sourced exclusively from Binance. This decision was driven by practical considerations: the altcoin pairs strategy targets high-frequency mean reversion (median holding period ∼6 hours) where execution quality and cost are paramount. DEX execution costs on Ethereum mainnet (0.50%–1.50% all-in per trade) would consume the strategy's edge entirely, and even on cheaper L2 chains, the 0.30%–0.60% round-trip cost and MEV exposure would materially impair performance.

I evaluated DEX data (GeckoTerminal, Uniswap subgraphs) during the data collection phase but deprioritised it due to lower liquidity, wash trading concerns, and the overhead of multi-chain data reconciliation.

The universe is constructed using a sticky top-100 approach via `binance_top100_15m.py`. Each month, the top 100 USDT-quoted perpetual pairs by 7-day average daily quote volume are selected. Coins that drop out of the top 100 but maintain above $2M daily volume are carried over as "hold" symbols, only being dropped when volume falls below this threshold or the pair is delisted from Binance.

Over the full January 2022 to December 2025 period, 405 unique symbols entered the universe, with approximately 140 active in any given month and roughly 10–20 fresh entrants each month.

Before screening, the universe undergoes an exclusion filter that removes stablecoins (USDT, USDC, DAI, BUSD, and 17 others), BTC wrappers (WBTC, BTCB, RENBTC, etc.), and ETH liquid staking derivatives (stETH, wstETH, cbETH, etc.). These pegged assets would produce spurious cointegration results, they are cointegrated by construction, not by market dynamics, and their spreads are driven by peg mechanisms rather than mean-reverting economic relationships.

Deviation from case study specification:

The case study specifies sector-based pair construction (L1, L2, DeFi, gaming, etc.) with intra-sector screening. Our implementation abandons sector constraints entirely: every non-excluded coin is tested against every other non-excluded coin in a single "all" sector. Although I built a comprehensive sector mapping (42 categories covering all 405 symbols), I found that imposing sector constraints was suboptimal for two reasons.

First, many of the strongest cointegration relationships in crypto occur across traditional sector boundaries: for example, an L1 token may be more tightly cointegrated with a DeFi token deployed on its chain than with another L1. Second, the Engle–Granger test with restrictive p-values and the subsequent filtering pipeline already reject pairs that lack genuine statistical relationships, making sector pre-filtering redundant. Allowing all-versus-all screening broadens the opportunity set without compromising signal quality.

4.2 Multi-Stage Screening Pipeline [2]

Pair selection is performed every 720 bars (approximately 7.5 days) using a fail-fast pipeline that orders tests from cheapest to most expensive, ensuring that the majority of the roughly

20,000 candidate pairs tested at each refit are rejected by inexpensive operations before reaching the computationally intensive later stages.

4.2.1 Stage 1 – Return Correlation Pre-Filter

A vectorised correlation matrix is computed on log-returns over the screening lookback window (3,000 bars, approximately 31 days). Pairs with return correlation below 0.40 are immediately rejected. This eliminates the vast majority of candidates before any regression is performed.

4.2.2 Stage 2 – Engle-Granger Cointegration Test

Surviving pairs are tested for cointegration using a fast NumPy-based implementation of the Augmented Dickey–Fuller test (bypassing `statsmodels`' full OLS model construction for a 5–10x speedup). The p-value threshold is set at 0.001, deliberately restrictive to counteract the multiple testing burden inherent in screening ∼20,000 pairs from a ∼140-coin universe.

Pairs with negative hedge ratios are dropped. While technically cointegrated, a negative beta indicates the assets move in opposite directions, and a pairs trade on such a pair leaves the portfolio directionally exposed rather than market-neutral.

4.2.3 Stage 3 – Half-life Filter

The half-life of mean reversion is estimated via an AR(1) model on the Engle–Granger residuals. Pairs with half-lives outside the tradeable range of 1 to 480 bars (15 minutes to approximately 5 days) are rejected. Within this range, the pair ranking algorithm (described below) assigns optimal scores to half-lives between 12 and 64 bars (3–16 hours), reflecting the strategy's target holding period.

4.2.4 Stage 4 – ADF + KPSS Conjunction Stationarity Test

Both the Augmented Dickey–Fuller test (which must reject the unit root null, $p < 0.10$) and the KPSS test (which must fail to reject the stationarity null, $p > 0.05$) must agree that the spread is stationary. This conjunction approach provides stronger evidence than either test alone and is robust to the conflicting type I/type II error profiles of the two tests.

4.2.5 Stage 5 – Variance Ratio Test

The variance ratio is computed at lags 2, 5, and 10 and must be below 1.0 at all lags. For a mean-reverting process, variance should grow sub-linearly with time (moves are being reversed), producing a ratio below 1.0. A ratio near or above 1.0 indicates random walk behaviour, the spread may pass cointegration tests by chance but lacks the reverting dynamics needed for profitable trading.

4.2.6 Stage 6 – Spread Distribution Filter

The spread's skewness and excess kurtosis are checked, with rejection thresholds at $|\text{skew}| > 1.0$ and excess kurtosis > 5.0. The z-score signal relies on the spread being roughly symmetric and approximately normal; highly skewed or fat-tailed spreads would produce misleading z-score readings and unreliable entry/exit signals.

4.2.7 Stage 7 – Structural Break Detection

A Block–Pelt change-in-mean detector (Bai–Perron style) is applied to block-aggregated spread means. Raw OU process samples are heavily autocorrelated, causing standard change-point algorithms to detect every oscillation as a "break." By aggregating into blocks of approximately one half-life, the block means become approximately i.i.d., allowing the Pelt algorithm to function correctly. Pairs with any detected breakpoint in the most recent 25% of the window are rejected, as a recent structural break indicates the cointegration relationship may be changing.

4.2.8 Stage 8 – Benjamini-Hochberg FDR Correction

All surviving pairs' p-values are subjected to a Benjamini–Hochberg correction at $q = 0.001$. This means that of every 1,000 pairs the pipeline selects, I accept at most 1 false positive. This is essential given the scale of testing: without FDR correction, the 0.001 per-test p-value would still produce roughly 20 false positives from 20,000 tests. With BH correction, the expected false discovery rate across the entire selected set is controlled at 0.1%.

4.2.9 Pair Ranking and Selection

Pairs surviving all eight stages are ranked by a weighted composite score combining: p-value (weight 1.0, smaller is better), half-life optimality (weight 1.5, score of 1.0 for half-lives in the 12–64 bar optimal range with exponential decay outside), variance ratio (weight 1.5, lower is better), return correlation (weight 1.0, higher is better), skewness (weight 0.5, closer to zero is better), and number of recent structural breaks (weight 0.5, fewer is better).

From the ranked list, the top 30 pairs are selected subject to a diversification constraint: any individual coin may appear in at most 3 different active pairs. This constraint serves two purposes. First, it prevents the portfolio from becoming overexposed to a single asset's idiosyncratic risk, if ETH crashes and is present in 10 pairs, all 10 would be affected simultaneously. Second, it forces the selection algorithm to reach beyond the large-cap coins that tend to be cointegrated with many partners, pushing the portfolio toward a more diverse set of pair relationships.

4.3 Signal Generation and Kalman Filter

4.3.1 Kalman Filter for Hedge Ratio Estimation

Deviation from case study specification:

The case study baseline specifies OLS regression for hedge ratio estimation. Our implementation replaces OLS entirely with a Kalman filter [1], following the approach documented in the academic literature as superior for dynamic hedge ratio tracking in pairs trading.

The Kalman filter models the spread between two log-price series as:

$$y_t = \alpha_t + \beta_t \cdot x_t + \epsilon_t$$

where α_t (intercept) and β_t (hedge ratio) are treated as time-varying state variables that evolve according to a random walk. The filter's process noise parameter δ (which governs how quickly the state can change) is fitted independently for each pair via maximum likelihood estimation, allowing the filter to automatically adapt to each pair's dynamics; slowly varying

pairs receive a small δ and thus a stable hedge ratio, while rapidly evolving pairs receive a larger δ and a more responsive filter.

The advantages over rolling OLS are substantial. The Kalman filter automatically handles the weighting of recent versus older data, eliminating the need to choose a fixed window length, a choice that is always a compromise between responsiveness and stability. It produces a posterior uncertainty measure (P_{11}, the posterior variance of the hedge ratio) as a natural byproduct, which I use directly in position sizing. It is computationally $\mathcal{O}(1)$ per bar update versus $\mathcal{O}(n)$ for a rolling OLS regression, and I further accelerate it via Numba JIT compilation and batch processing across all active pairs simultaneously. In our testing, the Kalman filter also produces more stable hedge ratios than OLS, rolling OLS estimates jump at each new data point as the window advances, while the Kalman filter's Bayesian update produces smoother transitions.

The filter is initialised with a warmup period of 1,000 bars (approximately 10 days) at each refit, during which it processes historical data to establish stable state estimates before generating live signals.

4.3.2 EWMA Z-Score Signal

The primary trading signal is an exponentially weighted moving average z-score of the Kalman spread, computed over a rolling window of 2,000 bars with a minimum warmup of 50 bars. The EWMA approach was adopted to address a specific problem observed with simple rolling z-scores: sudden spikes in the spread (caused by, for example, a flash crash in one leg of the pair) would pollute the mean and standard deviation estimates for days afterwards, producing distorted signals long after the event.

The EWMA half-life is set as a function of each pair's mean-reversion half-life. Pairs with faster reversion rates have shorter EWMA windows and thus more responsive z-scores, while slower-reverting pairs use longer windows that smooth over transient noise. This adaptive approach ensures that the signal sensitivity is matched to the underlying dynamics of each pair.

Entry is triggered when the absolute z-score exceeds 2.0, and the signal-based exit occurs when the z-score reverts to below 0.75 [3]. The entry threshold of 2.0 provides a balance between signal frequency and profitability. Lower thresholds generate more trades but with smaller expected profit per trade, while higher thresholds reduce frequency to the point where the strategy sits idle for extended periods. The exit at 0.75 rather than 0 is supported by Leung and Li (2015), who show that optimal mean-reversion trading with transaction costs should exit before full reversion to account for the cost of the trade and the risk of the spread re-widening.

4.4 Position Sizing and Risk Management

4.4.1 Volatility-Adjusted Leg Sizing

Rather than using the hedge ratio directly to determine dollar amounts for each leg, the strategy employs variance-neutral sizing: the two legs are sized so that each contributes equal variance to the combined position. Given a total notional allocation and a hedge ratio β with per-bar volatilities σ_A and σ_B estimated by the EWMA volatility tracker, the notional for each leg is computed so that $N_A \cdot \sigma_A = N_B \cdot \sigma_B$. This approach prevents the higher volatility leg from dominating the position's P&L, which is particularly important in crypto markets where

volatility can differ dramatically between a large-cap L1 token and a smaller DeFi protocol token. A safety valve caps the maximum leg imbalance at 5 : 1 to prevent extreme allocations when volatility estimates are noisy.

4.4.2 Implied Correlation Regime Filter

I introduce an *implied correlation* metric ($\rho_{implied}$) derived by decomposing the Kalman hedge ratio into its constituent parts. Since $\beta = \rho \times (\sigma_A/\sigma_B)$, I can recover the implied correlation as $\rho_{implied} = \beta \times (\sigma_B/\sigma_A)$. This decomposition reveals why the hedge ratio is what it is. When implied correlation is high (0.7–1.0), the pair relationship is driven by genuine co-movement and the mean-reversion thesis is sound. When implied correlation is low (< 0.5), the hedge ratio is being explained by volatility differences rather than correlation. The pair is degrading and trades should be avoided or downsized.

The implied correlation feeds into a smooth regime confidence scalar that modulates position size. At $\rho_{implied}$ between 0.8 and 1.0, confidence is at 1.0 (full sizing). Between 0.25 and 0.8, confidence linearly ramps from 0 to 1. Below 0.1, the trade is blocked entirely. Above 1.2, the trade is also blocked. A correlation above 1.0 is theoretically impossible and indicates that the Kalman beta and EWMA volatility estimates are disagreeing, signalling model stress.

4.4.3 Confidence-Based Sizing

Final position size integrates three independent signals: the Kalman filter's posterior uncertainty (P_{11}), the regime confidence from implied correlation, and a base allocation of 5% of equity per pair. These are combined multiplicatively with a confidence power parameter of 3.5, which sharpens the distinction between high confidence and low-confidence trades. A pair with 80% confidence is sized at approximately $0.8^{3.5} \approx 0.46$ of full allocation, while a pair at 50% confidence receives only $0.5^{3.5} \approx 0.09$ of full allocation. This aggressive confidence scaling means the strategy deploys most of its capital in its highest-conviction trades.

4.4.4 Exit Cascade

Open positions are evaluated through a priority-ordered exit cascade at each bar:

1. **Z-score hard stop ($z > 6.0$):** An extreme z-score indicates a crisis event in one of the pair's constituents. The position is closed immediately regardless of P&L.

2. **P&L stop-loss (-10%):** If the position has lost 10% or more of its entry notional, it is closed to prevent further deterioration. This catches cases where the pair has structurally decoupled.

3. **Maximum hold time ($6\times$ half-life):** If the position has been held for six times the pair's estimated half-life, it is exited. Under the OU model, the probability of not reverting within 6 half-lives is approximately $2^{-6} \approx 1.6\%$. A duration this long strongly suggests the cointegration relationship has changed.

4. **Beta break ($> 10\%$ drift):** If the Kalman hedge ratio has drifted more than 10% from its value at entry, the position is closed and the pair is placed on a 7-day cooldown. Large hedge ratio changes indicate a structural shift in the pair's dynamics.

5. **Signal exit ($z < 0.75$):** The spread has reverted sufficiently toward the mean. This is the desired profitable exit.

6. **Hedge ratio rebalance (> 5% drift):** If the hedge ratio has drifted by more than 5% but less than 10%, the position is rebalanced by trading only the delta required to adjust leg sizes to the new ratio. Rebalancing is limited to at most once per day to avoid excessive turnover.

Additionally, if a coin's data goes null (indicating a delisting or data outage), the position is force-closed on the last valid bar. If a pair is crowded out of the top 30 at a refit, replaced by a better-scoring pair, the position is closed.

4.4.5 Additional Risk Controls

A pair cooldown of 672 bars (approximately 7 days) is applied after any stop-loss exit or beta-break exit, preventing the strategy from immediately re-entering a pair that has demonstrated structural instability. A beta stability filter checks the coefficient of variation of hedge ratios across recent refits, blocking trades on pairs where the relationship is too volatile over time. Finally, the portfolio enforces a per-coin appearance limit of 3 pairs, preventing concentration in any single asset.

4.5 Strategy Performance

4.5.1 Headline Results

The strategy was backtested from January 2022 to December 2025 on $1,000,000 initial capital with 3× maximum leverage (Binance cross-margin). All results below are net of transaction costs ($1,555,434 cumulative) and margin interest ($4,361).

Table 1: Altcoin Pairs Strategy Performance Summary

Metric	Value
total_return	123.7943%
annualized_return	22.3110%
annualized_vol	12.00%
sharpe_ratio	1.446
sortino_ratio	1.624
calmar_ratio	1.971
max_drawdown	-11.3197%
avg_drawdown	-2.6137%
num_trades	6690
win_rate	61.0314%
avg_win	1978.85
avg_loss	-2537.24
profit_factor	1.221
avg_trade_pnl	218.99
total_costs	1555433.68
cost_drag	155.5434%

Continued on next page

Table 1 continued from previous page

Metric	Value
avg_bars_held	32.13
median_bars_held	25.00
max_leverage_setting	3.00
liquidation_events	0.00
peak_margin_usage	0.55
peak_actual_leverage	1.65
total_interest_paid	4360.52

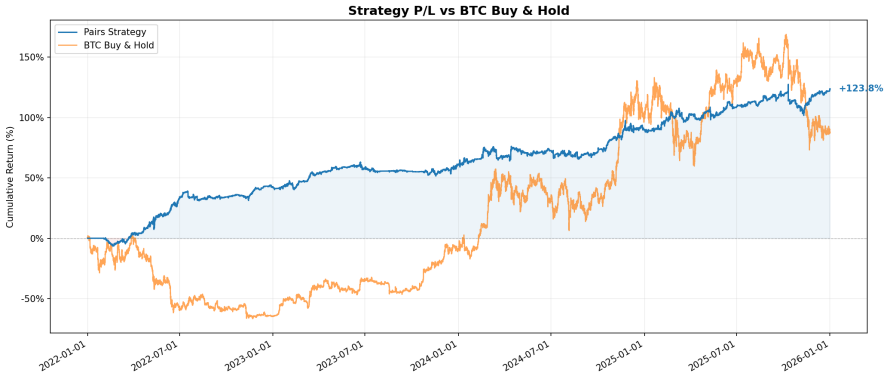

Figure 1: Strategy P/L vs BTC Buy & Hold

The equity curve shows steady, roughly linear growth with low correlation to BTC buy-and-hold. While BTC experienced drawdowns exceeding 50% during the period (notably during the UST/Luna collapse and FTX bankruptcy), the pairs strategy maintained positive returns through all four major crisis events. The only significant drawdown of approximately 10% occurred during the November 2025 BTC liquidation event, from which the strategy recovered within weeks.

4.5.2 Trade Quality Analysis

The distribution of per-trade net P&L reveals a clear positive edge, with returns centred around +0.38% per trade (median: +0.380%, mean: +0.378%). This consistency is the strategy's defining characteristic: rather than relying on large occasional wins, it extracts a small, repeatable edge across thousands of trades.

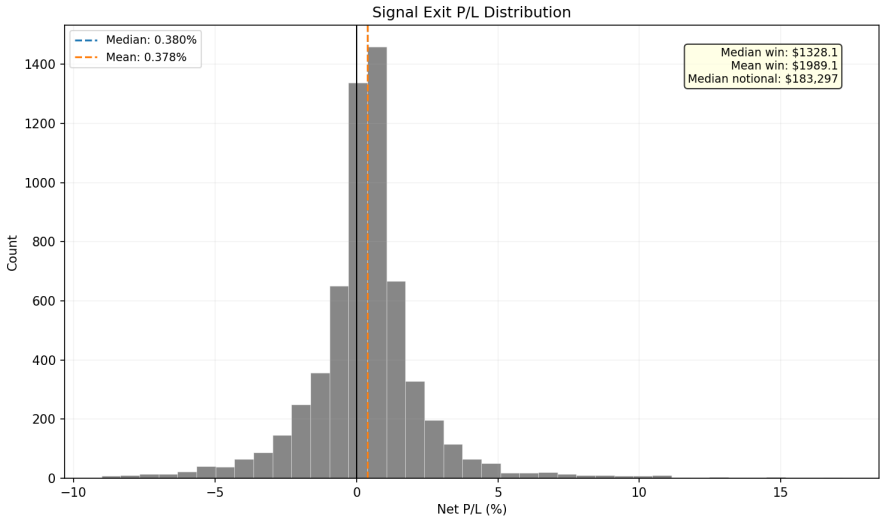

Figure 2: Signal Exit P/L Distribution

Comparison of the top 25% and bottom 25% of trades (ranked by P&L) shows that the quality metrics function as intended: winning trades have approximately 10–20% higher values across entry implied correlation (0.78 vs 0.71), regime confidence (0.87 vs 0.76), and overall entry quality score (0.66 vs 0.56). Winning trades also tend to be held for shorter durations (24 bars vs 47 bars), confirming that the mean-reversion thesis works best for pairs that revert quickly. Average notional on winning trades is larger than on losing trades ($218K vs $182K), while average notional on losing trades is about 15% smaller, both consequences of the confidence-based sizing system allocating more capital to higher-quality signals.

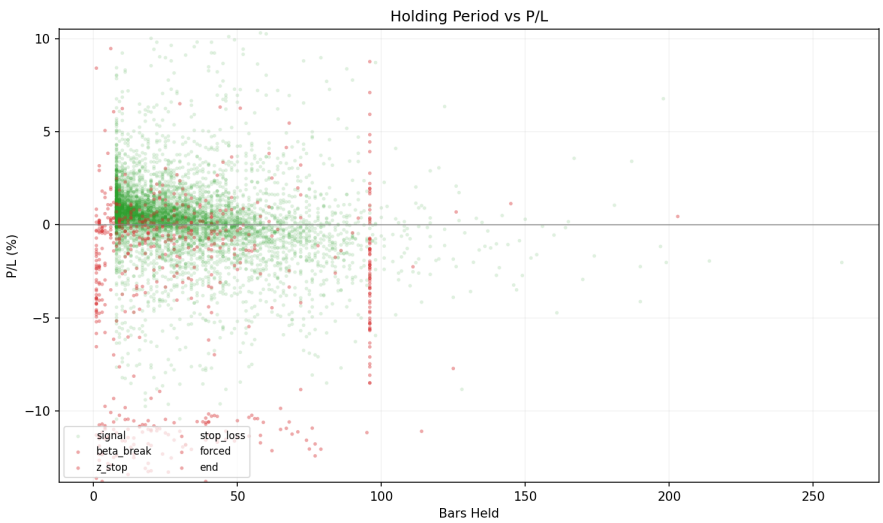

Figure 3: Holding Period vs P/L Scatter

4.5.3 Rolling Sharpe Analysis

The 90-day rolling Sharpe ratio exhibits clear regime shifts, oscillating between periods of strong performance (Sharpe > 3) and occasional negative periods. The overall Sharpe of 1.45 is the time-weighted average across these regimes. Notably, the strategy tends to perform best during sideways ("crab") market conditions where correlation structures are stable and

mean-reversion opportunities are abundant, and worst during violent directional moves where correlations spike and pairs temporarily decouple. This observation suggests that augmenting the strategy with explicit regime detection (e.g. an HMM on BTC volatility or cross-sectional correlation levels) could meaningfully improve risk-adjusted returns by reducing position sizes during hostile regimes.

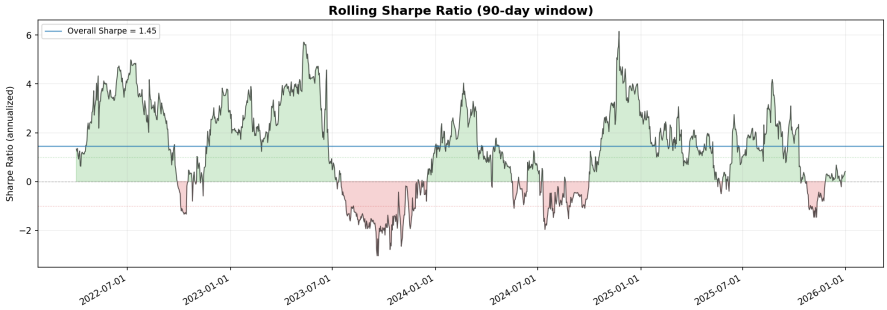

Figure 4: Rolling Sharpe Ratio (90-day window)

4.6 Crisis Performance Analysis

The strategy was evaluated across four major crypto market events to assess resilience under stress.

4.6.1 UST/Luna Collapse (May–June 2022)

The UST/Luna collapse triggered the largest single drawdown event in DeFi history, with contagion spreading across the entire crypto market as Luna's role as collateral in numerous protocols unwound. During this period, the strategy executed 383 trades with a 67.6% win rate, generating $169,082 in net P&L which is substantially above the strategy's average daily earnings. The high win rate and elevated P&L suggest that the crisis created widespread temporary dislocations between cointegrated pairs as panic selling hit some tokens faster than others, providing abundant mean-reversion opportunities. Only 9 trades (2.3%) were stopped out, with 350 of the 383 trades (91.4%) exiting on the normal signal.

4.6.2 FTX Bankruptcy (November–December 2022)

The FTX collapse and associated CEX trust crisis produced 186 trades at a 68.8% win rate and $87,806 in P&L. As with the Luna event, the win rate was well above the strategy's overall average (61%), suggesting that exchange-level crises generate pair dislocations that subsequently revert. Only 3 stop-losses were triggered, and the average holding period shortened to 25 bars, indicating faster mean reversion during the acute phase of the crisis.

4.6.3 USDC Depeg / Banking Crisis (March 2023)

The Silicon Valley Bank failure and temporary USDC depeg produced 134 trades at a 64.9% win rate and $71,618 in P&L. This event was less volatile for altcoin pairs specifically (the depeg primarily affected stablecoins, which are excluded from our universe), though the broader risk-off sentiment still created tradeable dislocations. Exit reasons were dominated by signal exits (122 of 134, or 91%), indicating orderly mean reversion.

4.6.4 SEC Lawsuits – Binance and Coinbase (June–July 2023)

The SEC enforcement actions represented a different type of stress: a regulatory event that selectively impacted certain tokens (those classified as securities) while leaving others unaffected. This produced 209 trades but a lower win rate of 59.3% and more modest P&L of $13,929. The average holding period lengthened to 36 bars and the number of beta-break exits increased (4 vs 1–3 in other crises), suggesting that the regulatory action caused more persistent structural changes in pair relationships that the strategy had difficulty distinguishing from temporary dislocations. This event also showed the highest proportion of forced exits (5) and stop-losses (5), confirming that regulatory events are more challenging for a pure statistical arbitrage strategy than market-driven crises.

4.6.5 Summary

The strategy was profitable through all four crisis events, though the nature of the crisis matters. Market-driven panic events (Luna, FTX, banking crisis) tend to produce abundant, fast-reverting dislocations that benefit the strategy, while regulatory events produce more structural pair breaks and lower profitability. In no crisis did the strategy experience a drawdown exceeding its overall maximum of 11.3%.

4.7 Capacity and Sensitivity

4.7.1 Capacity Estimate

A rigorous capacity analysis would require order-book data, which is not included in the current dataset. As a heuristic estimate based on daily volume: the lowest-volume coins in the active universe trade at $5–$10M per day. Staying under 1–2% of daily volume to avoid market impact, the strategy should support approximately $3M in deployed capital without difficulty. Optimising for capacity, spreading capital across more pairs, incorporating order-book data and liquidity filters directly into the screening pipeline, and avoiding the lowest-volume pairs could extend capacity to $10M or more.

4.7.2 Parameter Sensitivity

The strategy is sensitive to two key parameters:

- **Maximum pairs a coin can appear in (`max_coin_appearances = 3`):** This constraint controls the balance between large-cap and long-tail exposure. Large-cap coins (ETH, BNB, SOL) tend to be cointegrated with many partners, so a high limit would concentrate the portfolio in pairs involving these few assets. Lowering the limit forces the selection algorithm toward smaller, less correlated pairs, improving returns but also increasing volatility and reducing capacity.
- **Maximum concurrent pairs (`max_open = 30`):** Increasing the number of simultaneous pairs reduces portfolio volatility through diversification but dilutes returns, as the ranking system ensures that optimal pairs are selected first and marginal additions are lower quality. Decreasing it concentrates capital in fewer, higher-conviction pairs with higher returns but greater drawdown risk.

4.8 Deviations from Case Study Specification

Table 2: Implementation Deviations from Case Study

Specification Requirement	**Our Implementation**	**Rationale**
Dual-Uiverse (CEX + DEX)	CEX-only (Binance)	DEX execution costs (0.5–1.5% round-trip) exceed the strategy's per-trade edge. MEV exposure adds further risk.
Sector-based pair construction	All-versus-all screening	Strongest pairs often cross sector boundaries. The statistical pipeline already filters non-genuine relationships.
OLS Regression for Hedge Ratio	Kalman filter with MLE-fitted process noise	Kalman is well-documented as superior: adaptive weighting, built-in uncertainty, O(1) per bar, more stable estimates.
Standard z-Score	EWMA z-score (half-life adaptive)	Addresses spike-pollution problem in rolling z-scores. EWMA window auto-scales to each pair's reversion speed.
CEX vs DEX Performance Comparison	Not applicable (CEX only)	Follows from the CEX-only universe decision.
Walk-forward Optimisation	Continuous rolling backtest	The 7.5-day refit with FDR-controlled screening provides implicit walk-forward validation.

4.9 Enhancements Beyond Specification

Table 3: Enhancements Beyond Specification

Enhancement	**Description**	**Impact**
Implied correlation regime filter	Decomposes the Kalman hedge ratio β into implied correlation ρ and volatility ratio (σ_A/σ_B) using $\rho = \beta \times (\sigma_B/\sigma_A)$. Trades are blocked when ρ falls below 0.1 (pair degrading) or exceeds 1.2 (model stress). Between 0.25 and 0.8, position size ramps linearly.	Prevents entry on pairs where the hedge ratio is explained by volatility differences rather than genuine co-movement. Top-quartile trades show ∼10% higher implied ρ than bottom-quartile trades (0.78 vs 0.71), confirming the metric's discriminative power.
Volatility-adjusted leg sizing	Each leg is sized so that its dollar-volatility contribution is equal: $N_A \cdot \sigma_A = N_B \cdot \sigma_B$, using 7-day EWMA per-bar volatility estimates. A $5:1$ maximum imbalance cap prevents extreme allocations when vol estimates are noisy.	Ensures the spread P&L is balanced across legs. Without this, a pair of a high-vol small-cap (∼80% annualised) against a low-vol large-cap (∼40% annualised) would have the small-cap leg dominating position outcomes, undermining the market-neutral thesis.
Multi-signal confidence-based sizing	Final position size = base allocation × (Kalman P_{11} confidence) × (regime confidence)$^{3.5}$. The power parameter of 3.5 sharpens the distinction: 80% confidence → ∼46% of full size; 50% confidence → ∼9% of full size.	Concentrates capital in highest-conviction trades. Winning trades average 75% entry confidence versus 71% for losers, and winning trades are sized ∼20% larger in notional ($218K vs $182K), showing the system allocates more to better opportunities.
Block-Pelt structural break detection	Aggregates the OU spread into non-overlapping blocks of one half-life, computes block means (absorbing within-block autocorrelation), then runs Pelt change-in-mean detection. Rejects pairs with any breakpoint in the most recent 25% of the screening window.	Standard change-point algorithms (e.g. raw Pelt, CUSUM) produce > 90% false positive rates on autocorrelated OU processes. Block aggregation reduces this to ∼4% while maintaining 94% detection of 3σ regime shifts. Applied last in the pipeline since it is the most computationally expensive filter.

Continued on next page

Table 3 continued from previous page

Specification Requirement	Our Implementation	Rationale
Pair cooldown system	After any stop-loss, z-hard-stop, or beta-break exit, the pair is blacklisted for 672 bars (∼7 days). During cooldown, no new entries are permitted on that pair regardless of signal strength.	Prevents the strategy from re-entering a pair that has just demonstrated structural instability. Without cooldown, the screening pipeline might re-select the pair at the next refit while the underlying dislocation is still unresolved, leading to repeated stop-outs on the same pair.
Full cross-margin simulation	Models Binance spot cross-margin: position-level initial and maintenance margin tracking, hourly interest charges on borrowed amounts (0.03%/hour), forced liquidation when equity breaches maintenance margin, and a 2% liquidation fee. Margin state is snapshotted alongside equity at each recording interval.	Produces realistic leveraged returns rather than assuming unlimited borrowing capacity. Despite 3× maximum leverage, peak actual leverage reached only 1.65× with zero liquidation events, demonstrating that the confidence-based sizing naturally self-regulates exposure. Total interest cost was $4,361—negligible relative to $1.46M gross P&L.

4.10 Future Improvements

4.10.1 Strategy Enhancements

The backtesting framework and screening pipeline are mature, but several targeted improvements would materially improve live performance.

Cost-aware entry gating: The strategy currently enters any trade that passes the screening pipeline and z-score threshold, regardless of whether the expected profit justifies the transaction cost. A natural extension is to estimate the expected P&L for each candidate trade using the pair's half-life, current z-score distance to the exit threshold, and historical spread volatility, and gate entries where the predicted cost exceeds the profit opportunity. Given that transaction costs totalled $1.56M over the backtest (a cost drag of 155% on initial capital), even a modest improvement in trade selection efficiency would have a large absolute impact on net returns.

Regime-aware position sizing: The rolling Sharpe analysis (Section 4.5.3) reveals clear regime shifts, with periods of sustained positive performance alternating with drawdown episodes. The strategy currently applies the same sizing logic regardless of the broader market environment. Adding an explicit regime detector, for example, an HMM on cross-sectional

correlation levels or BTC realised volatility would allow the strategy to scale down exposure during hostile regimes (high correlation, directional markets) and lean in during favourable ones (low correlation, range-bound markets). This is the single most promising improvement for risk-adjusted returns.

Dynamic capital allocation: The current sizing model allocates a fixed 5% of equity per pair regardless of the pair's liquidity or expected capacity. This is inefficient in two ways: high-volume pairs that could absorb larger positions are undersized, while the portfolio's aggregate exposure often utilises only roughly half of available capital, effectively forcing inflated leverage settings to deploy a reasonable fraction of equity. A smarter allocation model would route more capital to higher-volume pairs (where market impact is negligible) and less to illiquid ones, ensuring the portfolio is more fully deployed without increasing the leverage ceiling.

4.10.2 Venue Expansion

The single most impactful improvement would be expanding the universe beyond Binance to additional venues, particularly decentralised exchanges. Persistent pricing inefficiencies exist on less crowded venues, and these inefficiencies are even more pronounced for altcoin pairs, where DEX liquidity pools often price tokens differently from CEX order books due to different participant bases, AMM curve mechanics, and the absence of professional market makers.

In hindsight, the data collection effort was disproportionately focused on building a sophisticated strategy for Binance, a highly competitive, efficient exchange where alpha is hardest to find. The real mispricings and structural inefficiencies lie on less crowded exchanges where competition is thinner and arbitrageur capital is scarce. Incorporating DEX price data (Uniswap, Raydium, Orca) and alternative CEX data (Bybit, OKX, MEXC) into the screening pipeline would broaden the opportunity set substantially. The main tradeoff is that these venues cannot support the same capital depth as Binance, so the strategy would need to be adapted for smaller position sizes and venue-specific execution costs but the wider spreads available should more than compensate.

References

[1] Daniel P. Palomar. *Kalman Filtering for Pairs Trading*, Section 15.6. Portfolio Optimization, 2024. `https://portfoliooptimization.com/`

[2] Nick Firoozye. Trading Mid-Frequency and RV like a Pro (Part 1): Identifying Mean-Reversion with Statistical Tests. *Objectively Random*, 2024. `https://objectivelyrandom.com/`

[3] Min Dai, Qing Zhang, and Qiji Zhu. Optimal Mean Reversion Trading with Transaction Costs and Stop-Loss Exit. *arXiv preprint arXiv:1411.5062*, 2014.

A Data Acquisition Plan

A.1 Data Requirements Matrix

Table 4: Data Requirements – Altcoin Statistical Arbitrage

Data Type	Asset Coverage	Timeframe	Frequency	Fields Required	Priority
Spot OHLCV	Top 100 USDT pairs by liquidity, re-evaluated monthly	2022–01 to 2025–12	15-minute	open, high, low, close, volume, quote_volume, trades, taker_buy_volume, taker_buy_quote_volume	Must-have
Universe Metadata	All active USDT spot pairs	Monthly snapshots	Monthly	symbol, baseAsset, rank, avg_daily_quote_volume, avg_daily_trades, status (top_n / carry_over / dropped)	Must-have
Market Cap Data	Same universe	Daily	Daily	market_cap, fully_diluted_valuation	Nice-to-have

A.2 Source Evaluation

Table 5: Selected Data Sources

Source	Quality Notes	Data Collected	Cost	API Limits	Type
Binance Spot API	Altcoin 15m OHLCV, universe ranking	Free	∼1200 req/min	Gold standard for CEX spot data; deepest liquidity for most USDT pairs	CEX

A.3 Decision Criteria

The primary criteria for source selection were: (1) data completeness across the full 2022–2025 window, (2) API reliability and speed, and (3) dominant market share for altcoin spot trading. Binance was chosen as the primary CEX source due to dominant market share and the availability of comprehensive historical kline data via its REST API.

A.4 Collection Strategy

A.4.1 Technical Approach

Script: `binance_top100_15m.py` – **Altcoin Universe + OHLCV**

This script uses the Binance REST API (`/api/v3/klines`, `/api/v3/exchangeInfo`) with a two-phase approach. In Phase 1 (Ranking), for each month, 7-day daily klines are fetched for all USDT pairs, ranked by average quote volume, with the top N selected and carry-over logic applied for continuity. In Phase 2 (OHLCV), 15-minute klines are fetched for the active universe, paginated at 1000 candles per request. The universe is "sticky": once a coin enters the top 100, it remains as long as it maintains more than $2M average daily volume. New entrants, carry-overs, and drops are tracked with status flags. Output consists of per-symbol per-month Parquet files, a `universe.csv`, and a `dropped.csv`.

A.4.2 Rate Limiting & Error Handling

The script implements exponential backoff on HTTP errors (429 rate limits, 5xx server errors), configurable rate limiting (Binance ∼8 req/s), retry logic with 5 retries and exponential wait (2^{attempt} seconds), and skip-on-error behaviour where pagination errors cause the script to skip forward (typically 7 days) rather than halt. The script also supports a `--skip-existing` flag to resume interrupted downloads, with universe ranking state persisted to CSV.

A.4.3 Storage Format

All outputs use Apache Parquet with the PyArrow engine, providing 10–50× compression versus CSV, columnar storage enabling efficient analytical queries, and native support for typed columns (timestamps, floats, categoricals).

A.4.4 Timeline and Risk Mitigation

Table 6: Collection Timeline and Risk Mitigation

Phase	**Estimated Duration**	**Risk**	**Mitigation**
Altcoin OHLCV (ranking + 15m klines)	2–4 days (API-bound)	Binance rate limiting or IP ban	0.12s delay between requests; 418 handler with 120s cooldown; `--skip-existing` for resume

A.4.5 Fallback Plans

Table 7: Fallback Plans by Data Source

Primary Source	**Failure Mode**	**Fallback**
Binance API (altcoins)	Rate limiting / IP ban	Reduce concurrency; spread collection across multiple days; use `--skip-existing`

B Data Quality Report

B.1 Coverage Analysis

Source: Binance Spot API via `binance_top100_15m.py`

Table 8: Altcoin Statistical Arbitrage Data Coverage

Dataset	Target Range	Actual Range	Frequency	Symbols	Rows	Status
Altcoin 15m OHLCV	2022–01 to 2025–12	2022–01 to 2025–12	15–minute	405	140,251	Complete
Universe Mapping	2022–01 to 2025–12	2022–01 to 2025–12	Monthly	∼140 per month	6,801	Complete

The altcoin dataset covers the full target range of January 2022 through December 2025 at 15-minute resolution. Across the collection period, 405 unique symbols entered the active universe, with approximately 140 symbols active in any given month. Of these, the top 100 were selected fresh each month by quote volume, with roughly 10–20 new entrants per month replacing symbols that fell below the carry-over threshold. The remaining ∼120–130 symbols each month were carry-overs from prior months that maintained sufficient volume. The universe mapping totals 6,801 symbol-month records.

The dataset comprises approximately 140,251 individual Parquet files (one per symbol per month), with minimal missing candles within covered ranges. Binance's spot API provides highly reliable kline data, and no significant gaps or outages were identified during the collection period.